TECHNICAL NOTICE

Third-party Runtime Components

MediaPipe provenance, modifications, checksums, licensing, and update procedure.

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The **PowerGlove Vision Controller (Arduino UNO Q)** hosts the camera and recognition runtime described below.

PowerGlove Vision's original source code and associated documentation are licensed under the repository's MIT License. That license does not replace the licenses or terms that apply to third-party software and model files.

MediaPipe 0.10.18 ARM64 wheel

A wheel (`.whl`) is an installable Python package. This wheel includes MediaPipe's compiled Linux ARM64 code, so the PowerGlove Vision Controller does not need to build it. The filename's `cp312-cp312` tags identify CPython 3.12 and its binary interface; `manylinux2014_aarch64` and `manylinux_2_17_aarch64` identify compatible ARM64 Linux environments. The PowerGlove Vision Controller uses this tracked file:

```
python/worker-wheels/mediapipe-0.10.18-cp312-cp312-manylinux2014_aarch64.manylinux_2_17_aarch64.whl
```

Property	Value
Component	MediaPipe 0.10.18 for CPython 3.12, Linux ARM64
Upstream project	<https://github.com/google-ai-edge/mediapipe>
Upstream package	<https://pypi.org/project/mediapipe/0.10.18/>
License	Apache License 2.0
PowerGlove repack SHA-256	`f2617c0960eb35aaa58b76076a9ae629edbf7ec8901d5fab4b046c28d13fbe8d`
Original PyPI wheel SHA-256	`09cbf7dc1f9a2deeaaac687e5f982836623def4cbd3e827d95f86f42450d2dd1`

Modification notice

PowerGlove Vision repackaged the upstream wheel for its headless PowerGlove Vision Controller worker. The Python package code and compiled MediaPipe binaries were not modified. The changes listed below were made when repackaging the wheel, and its `RECORD` file was rebuilt to reflect them:

- The `jax` dependency declaration was removed.
- The `jaxlib` dependency declaration was removed.
- The `opencv-contrib-python` dependency was replaced with `opencv-contrib-python-headless==4.10.0.84`.
- The upstream wheel's empty `mediapipe.libs` directory was omitted.

These changes avoid unnecessary JAX installation and GUI OpenCV dependencies on the PowerGlove Vision Controller. The repacked wheel retains MediaPipe's Apache 2.0 license at `mediapipe-0.10.18.dist-info/LICENSE`. Do not substitute the upstream wheel without retesting dependency resolution, camera startup, and hand tracking.

The released wheel and production Controller path are CPU-only. An isolated research build from the exact MediaPipe 0.10.18 source successfully initialized the UNO Q Adreno 702 through EGL/OpenGL ES and created a TensorFlow Lite GPU delegate. The first MediaPipe Tasks graph was substantially slower than the proven MediaPipe Hands path, so that custom wheel, its temporary Mesa alignment, and its build environment are not distributed. This confirms device access, not a production-quality GPU tracker. Any future lean GPU palm/landmark path must retain upstream Apache notices and pass the documented latency, continuity, recognition, jitter, and thermal gates before packaging.

Google Hand Landmarker model

PowerGlove Vision uses Google's float16 Hand Landmarker task bundle.

Property	Value
PowerGlove Vision Controller runtime path	`data/models/hand_landmarker.task`
Official download	<https://storage.googleapis.com/mediapipe-models/hand_landmarker/hand_landmarker/float16/1/hand_landmarker.task>
SHA-256	`fbc2a30080c3c557093b5ddfc334698132eb341044ccee322ccf8bcf3607cde1`
Size	7,819,105 bytes

The unmodified model is preserved in `models/hand_landmarker.task` and included in the App Lab installation ZIP. Its Apache 2.0 license is in `licenses/Apache-2.0.txt`; third-party notices record its source, checksum, and licensing evidence. Google's official Hand Landmarker documentation links a model card that explicitly states Apache License, Version 2.0 on page 2. The project's MIT license does not replace that license. No model modifications were made.

When vision is first activated, the application verifies and copies the bundled model into private, persistent `data/models/`. A verified cached copy is reused. A damaged cache is replaced from the bundle; a damaged bundle is rejected. Google's pinned download is used only when no bundled model is available. **Gestures off** does not open or install the model. Wi-Fi deployments preserve `data/` and include the bundled recovery copy. The model therefore does not require internet access in a complete installation package; Python and other first-install dependencies may still require downloads.

`scripts/fetch-runtime-assets.sh` follows the same bundled-first policy and verifies the pinned checksum before installing the private cached copy. `models/SHA256SUMS` records the preserved model's digest. Keep the model, license, notices, and checksum together in backups. Store a copy of the recovery archive on another drive or in your regular off-machine backup; copies on the same Mac do not protect against loss of that Mac.

Verified PowerGlove Vision Controller sketch toolchain

The September 4, 2026 sketch build and firmware deployment used these pins from `sketch/sketch.yaml`:

- Arduino Zephyr platform **1.0.0**
- Arduino_RouterBridge **0.4.3**
- Arduino_RPClite **0.3.0**
- ArxContainer **0.7.0**
- ArxTypeTraits **0.3.2**
- DebugLog **0.8.4**
- MsgPack **0.4.2**

The build resolved Arduino_LED_Matrix **0.1.3** from that platform. Preserve the dependency pins when synchronizing with App Lab; its shortened generated configuration is not a replacement for the project's complete configuration. Revalidate compilation and device operation before changing a pin.

Modified Nestopia libretro core

The optional `lr-nestopia-powerglove` core is built from libretro Nestopia revision `5a1cd378cb46ca9ccc2dd6f8b2b6a79ab986052e`. Nestopia identifies its license as the GNU General Public License, version 2. PowerGlove Vision's MIT license does not replace the license of Nestopia or the resulting modified core.

Property	Value
Component	libretro Nestopia with the PowerGlove Vision native-state patch
Upstream project	<https://github.com/libretro/nestopia>
Pinned revision	`5a1cd378cb46ca9ccc2dd6f8b2b6a79ab986052e`
Upstream license	GNU General Public License, version 2
Local modification	`native/nestopia-powerglove/nestopia-powerglove.patch`
Patch SHA-256	`3172ef337bfbb37c67ea2507544f21c7de3cedd25733802b062b0d02ef679397`
Modified upstream files	`libretro/libretro.cpp`; `source/core/input/NstInpPowerGlove.cpp`
Modification ledger	`native/nestopia-powerglove/CHANGES.md`
Build recipe	`scripts/build-nestopia-powerglove.sh`
Built core name	`nestopia_powerglove_libretro.so` on RetroPie
Installed core directory	`/opt/retropie/libretrocores/lr-nestopia-powerglove/`

Ordinary releases contain the patch and build recipe, not a compiled Nestopia core. If the user accepts the RetroPie installer's optional native-core step, the target machine downloads the pinned upstream source, including its author notices and `COPYING` file, applies the patch, and builds for its own processor. The core installer places `COPYING` and the local distribution note beside the installed binary. It also installs the separate PowerGlove Vision modification ledger. The original Nestopia copyright/GPL header in `NstInpPowerGlove.cpp` remains byte-for-byte intact, and the build stops if a future patch changes it. Stock Nestopia remains untouched and FCEUmm remains available.

If prebuilt cores are published in the future, produce separately identified artifacts for every tested RetroPie architecture. Accompany each binary with the exact complete corresponding source archive used to build it, the local patch and build instructions, all upstream notices, and the GPLv2 license. A Git commit or patch URL alone is not the project's binary-distribution plan. ROM images are never part of a source or binary core artifact.

At runtime, RetroArch loads the custom core only for an explicitly selected ROM. The launch entry passes the read-only latest-sample file through `POWERGLOVE_NATIVE_STATE`; the default path is `/run/powerglove/native-state`. The patch registers a separately named **Power Glove Vision** controller and identifies the library as **Nestopia PowerGlove**. Invalid, stale, uncalibrated, lost-tracking, or wrong-profile samples are neutralized. The compatibility record in Super Glove Ball native compatibility separates exact-ROM-confirmed X/Y/Z and hand-pose packet behavior from wrist and button fields that remain unmapped.

The local patch changes only `libretro/libretro.cpp` and `source/core/input/NstInpPowerGlove.cpp`. SHA-256 values for both pristine upstream files are recorded in the modification ledger. The build applies the patch only after checking out the pinned commit, rejects unrelated source-tree changes, and verifies that Nestopia's original 22-line Power Glove copyright and GPL header remains byte-for-byte identical.

External RetroPie emulator dependencies

PowerGlove Vision uses RetroPie-provided emulator software but does not include those binaries in its installation archives. When either dependency is absent, the RetroPie installer can ask the user's existing RetroPie Setup installation to install it. That operation remains governed by RetroPie and the upstream licenses.

Component	PowerGlove Vision use	Upstream and license	Distribution boundary
RetroArch	Libretro frontend used to load FCEUmm and `lr-nestopia-powerglove`	RetroArch, GPLv3	Installed by RetroPie; not modified or redistributed by PowerGlove Vision
FCEUmm	Default NES core for standard D-pad/button mappings and the complete Super Glove Ball fallback	FCEUmm, GPLv2	Stock RetroPie core; not modified or redistributed by PowerGlove Vision

The deterministic direction benchmark separately builds stock FCEUmm revision `236ccdfc911e84c60fea6b9d0699c2d440a8de14` in an isolated working directory. That pin makes the benchmark reproducible; it does not replace the user's RetroPie core, install FCEUmm, or make the benchmark binary a release artifact.

Updating runtime components

MediaPipe wheel or Hand Landmarker model

Before publishing a wheel or model update, complete these steps. The [command reference] explains the build and verification scripts.

1. Record the official source URL, version, license, size, and SHA-256 here.
2. Update the pinned values in `src/powerglove_vision/runtime_assets.py`, `scripts/fetch-runtime-assets.sh`, `scripts/verify-app-lab-package.py`, and `models/SHA256SUMS` when changing the model.
3. If repackaging another wheel, record every difference from upstream and retain its license files.
4. Build the App Lab installation ZIP and confirm it contains one wheel, the verified model, its license and notices, and only the root `sketch/` application sketch.
5. Test first-launch offline model installation, download fallback, and checksum verification, background preloading with capture off, first activation after reboot, camera initialization, tracking, the Glove Academy and Dashboard pages, and controller output on the PowerGlove Vision Controller before publishing the package.

Modified Nestopia core

Before changing the Nestopia revision or native patch:

1. Select an exact upstream commit from the official libretro Nestopia repository. Record the commit, upstream license, affected pristine-file SHA-256 values, and new patch SHA-256 in this document and `native/nestopia-powerglove/CHANGES.md`.
2. Update the identical revision pin in `scripts/build-nestopia-powerglove.sh`, the native-core README, the modification ledger, tests, and compatibility/benchmark documents. Do not use a moving branch or tag as the build identity.
3. Rebase `native/nestopia-powerglove/nestopia-powerglove.patch` onto a clean checkout. Preserve all upstream headers and notices. The guarded build must still reject changes to the original `NstInpPowerGlove.cpp` header.
4. Run the native-core, state-bridge, installer, selection, exact-ROM trace, safe-neutral, and direction-response tests. Reconfirm packet length, detection, bit order, boundaries, timing, X/Y/Z orientation, open/fist/index values, Start behavior, tracking-loss release, and the explicit FCEUmm rollback on the cabinet.
5. Build the RetroPie installation archive and verify it contains the patch, build/install recipes, modification ledger, and third-party notices, but no ROM or compiled core. If publishing a binary separately, provide the exact complete corresponding source and GPL materials described above.

When the benchmark FCEUmm pin changes, record the new official revision in the benchmark document and rerun both the native and standard-joypad lanes. Normal RetroPie FCEUmm and RetroArch upgrades remain the responsibility of RetroPie; retest controller selection and fallback gameplay before claiming compatibility.

Documentation and website assets

The following assets support the guides and browser interface. Their origins are recorded separately from the software, model, and firmware dependencies above.

Documentation illustration provenance

The gesture sheets under `docs/images/gestures/` were generated on September 3, 2026 with OpenAI's image-generation tool from project-authored prompts, then selected and arranged for the PowerGlove Vision gameplay guide. They are documentation assets, not runtime dependencies. No game screenshots, scans, box art, characters, publisher logos, or other source images were supplied to the generator.

The individual gestures and original Pixel Pal mascot under `docs/images/gestures/v2/` were generated on September 4, 2026 with the same built-in tool, using the project's generated contact sheet as a style reference. Their prompts are preserved in `docs/images/gestures/v2/prompts.json`. The earlier illustrations remain available in their original locations.

The index-curl illustration was subsequently redrawn using a user-supplied hand photograph as its pose reference. Only the illustrated glove is included; the reference photograph is not distributed with the project.

The repository applies its MIT License to these curated project assets to the extent the project owner has rights in them. Game names and other third-party marks remain the property of their respective owners.

Application screenshots

All application screenshots in `docs/images/` were refreshed from the current PowerGlove Vision source on September 6, 2026. They cover Dashboard, local Rock Paper Scissors, Glove Academy, personalization, players and hand-setup restoration, Setup and guided pairing, Games, and the Help library. `scripts/capture-guide-screenshots.py` renders the real page templates in an isolated browser with temporary player state and sample telemetry. Camera areas use an explicit "Camera preview omitted" placeholder; no live camera, cabinet, or personal settings are accessed. The script also invokes `tests/browser_setup_pairing.py --screenshots` for pairing states using non-secret fixtures. These captures document the interface, not live delivery or hardware verification. Gesture illustrations and physical matrix photographs remain unchanged. No runtime dependencies are added.

Website icon

The website icon in `assets/powerglove-vision-icon.png` was derived from the project's `assets/powerglove-vision-logo.png` on September 6, 2026 using OpenAI's image-generation tool. It isolates the hand-and-target emblem without the wordmark. Browser-tab and Apple touch icon variants were resized from that square artwork. `assets/favicon.ico` contains 16, 32, and 48 pixel variants and is the single browser-tab icon declared by the shared page template. The Apple touch icon is separate. Setup and Help use the same root-relative icon URL; there is no Setup-specific asset directory. Setup is served directly at `/setup`, without a query revision or redirect. Older query-string bookmarks still work. The original logo remains unchanged.